ABIGAIL JONES

abigailrjones.github.io
abigailjones@math.wisc.edu

EDUCATION

University of Wisconsin–Madison — Madison, Wisconsin

PhD in Mathematics (August 2023 – present)

Master's in Mathematics (August 2023 – December 2025)

GPA 4.00

Brigham Young University — Provo, Utah

BS in Mathematics (2018 – 2023)

GPA 3.98

CURRENT PROJECTS

Reference map technique (present)

Advisor: Chris Rycroft

Extending the reference map technique to modeling objects in fluids that bend enough to cause self contact, building a simulation engine in C++

Visiting as a Lawrence Berkeley Lab affiliate in June 2025, July 2026

Rigid homotopies (present)

Mentor: Jose Israel Rodriguez

Implementing a theoretical homotopy continuation algorithm in Julia, testing performance in practice and comparing to theoretical bounds

RESEARCH EXPERIENCE

R&D undergraduate summer intern (May 2023 – August 2023)

Mentor: Clifford Hansen

Sandia National Laboratories, FORCEE (Future of Research for Climate, Earth, and Energy)

Estimating parameters to convert between several models for the incidence angle modifier for photovoltaic arrays, contributing to the pvlib-python project

Undergraduate research assistant (October 2019 – April 2023)

Advisor: Michael Griffin

Studying Galois groups of torsion points on CM elliptic curves

Department number theory seminar

Norm–Euclidean project (March 2022 – April 2023)

Mentors: Michael Griffin, Nick Andersen

Writing and running code to identify candidate quadratic fields for the proof process given in Clark's "A Quadratic Field which is Euclidean but not Norm–Euclidean"

GitHub repository for Norm–Euclidean project

Technical undergraduate summer intern (May 2022 – August 2022)

Mentor: Clifford Hansen

Sandia National Laboratories, FORCEE (Future of Research for Climate, Earth, and Energy)

Modeling synthetic IV curves and filtering measured IV curves in Python, compiling measured IV curve data from Sandia databases for publication, presenting in FORCEE Student Symposium

GitHub repository for synthetic curves project

Final project in analytic number theory course (November 2021)

Replicating computations in Odlyzko and te Riele's "Disproof of the Mertens conjecture", writing an expository paper explaining this result for the course

GitHub repository for Mertens Conjecture project

TALKS

BYU Student Research Conference (March 2020 – 2023)

Won best undergraduate talk in session (2022)

FORCEE Student Symposium (August 2022)

Won best talk in session

TEACHING EXPERIENCE

University of Wisconsin–Madison

Calculus 3 (MATH 234) (Spring 2026 – Fall 2026)

In charge of two discussion sections twice a week, helping students with worksheets, proctoring and grading exams, holding weekly office hours

Calculus 2 (MATH 222) (Spring 2024 – Fall 2025)

Calculus 1 (MATH 221) (Fall 2023)

Brigham Young University

Calculus 2 (MATH 113) (Spring 2022)

In charge of classroom once a week, helping students with worksheets, writing review worksheets for tests, meeting one-on-one with students outside of class

Multivariable calculus (MATH 314) (Fall 2020)

Setting up course web page, grading assignments and exams, responding to student emails, proctoring exams

HONORS

Dean's List (2018 – 2020, 2022)
BYU Full scholarship (2018 – 2022)
Marine Corps Scholastic Excellence Award (2018)
National Merit Finalist (2018)

PROGRAMMING AND MARKUP LANGUAGES

Fluent: Python, C++, LaTeX
Conversational: SageMath, Julia
Tourist: HTML, CSS, Rocq, Lean